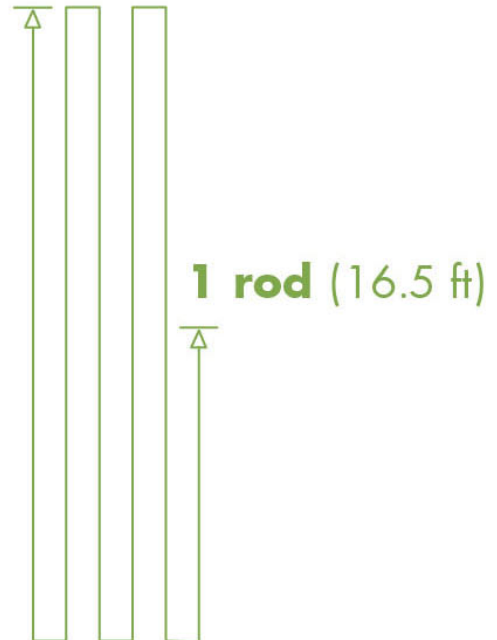
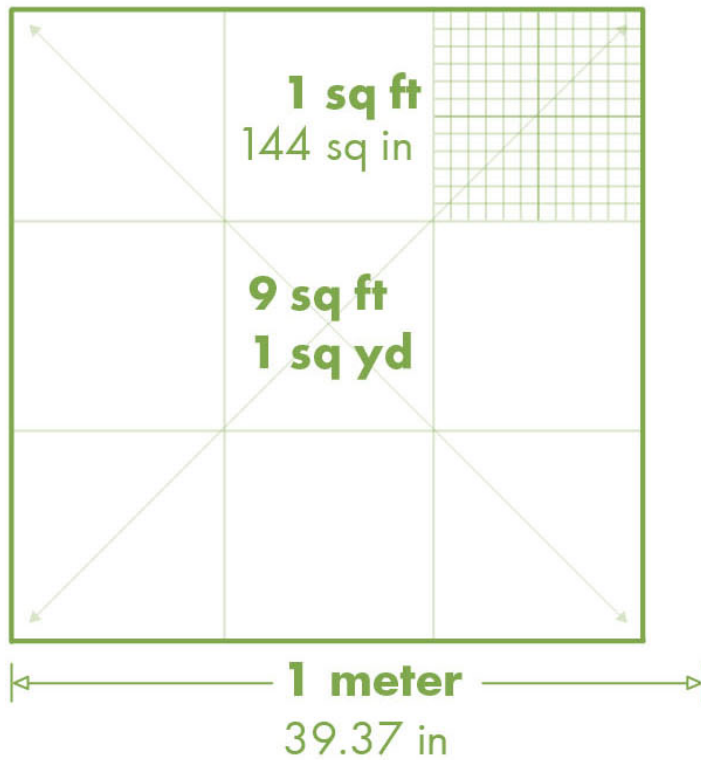
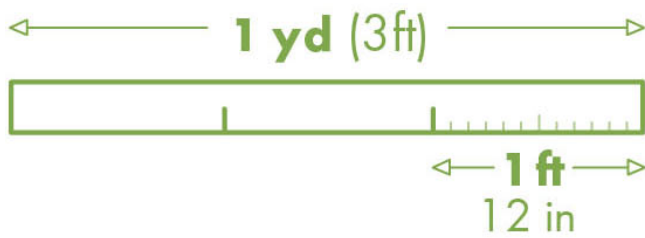


Front foot (frontage)

Length of property running along a street or highway



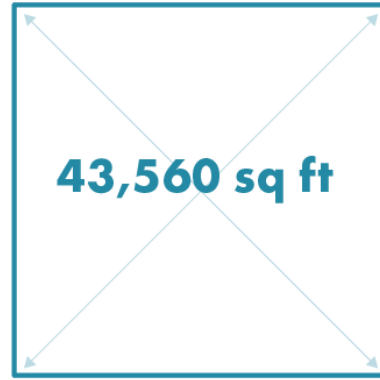


1 mile

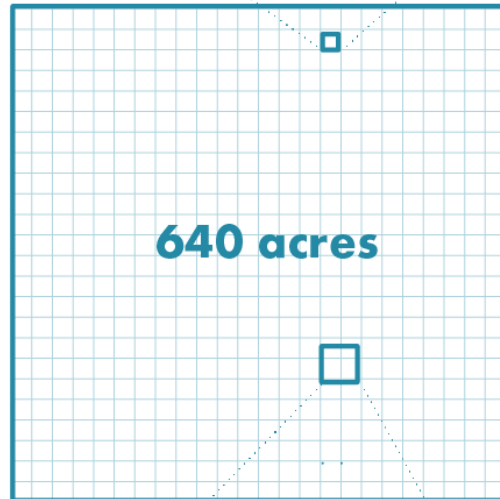


5,280 ft

=



1 acre



2.47 acres



Find Perimeter

■ **Square** = $4 \times \text{length}$

■ **Rectangle** = $(2 \times \text{length}) + (2 \times \text{width})$

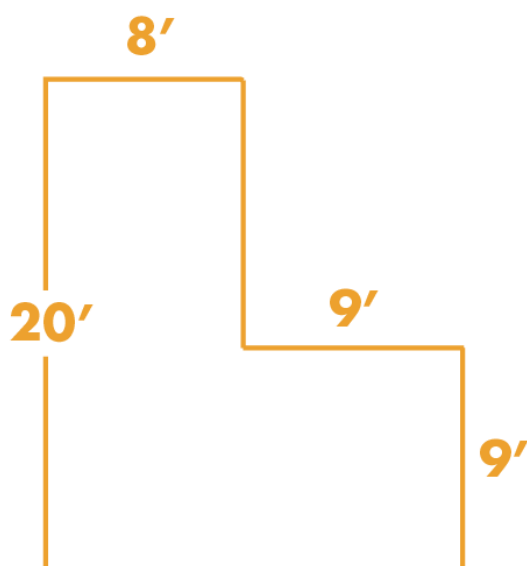
■ **Any other shapes** = **add lengths of each side**

Find the perimeter of a square

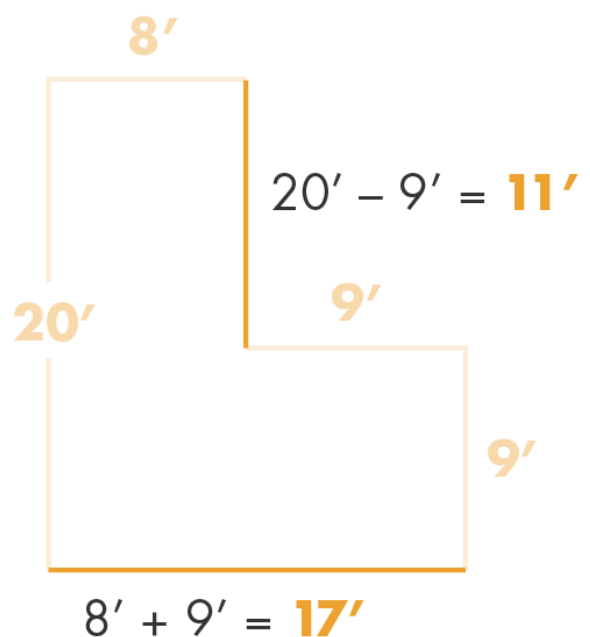


$$\begin{aligned}\text{Perimeter} &= 4 \times 14' \\ &= \mathbf{56'}\end{aligned}$$

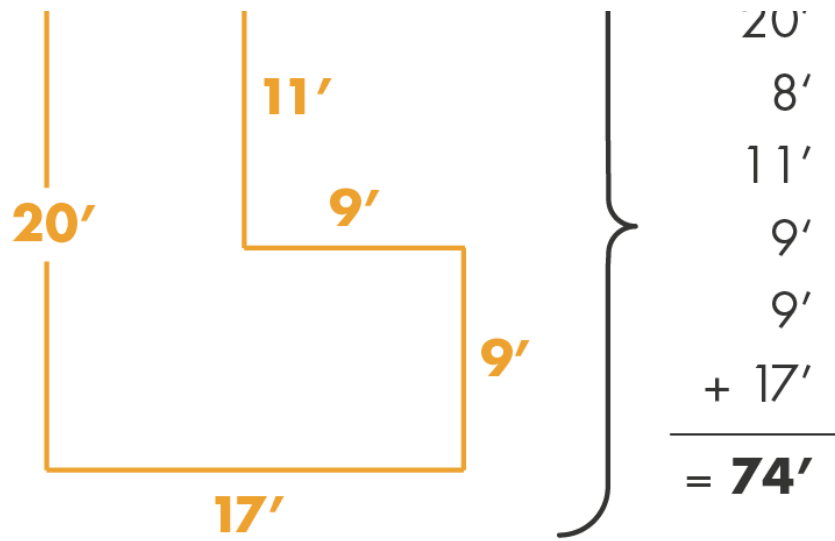
Find the perimeter of an irregular shape



>>



20'



Find Area

■ **Square = side × side**

■ **Rectangle = length × width**

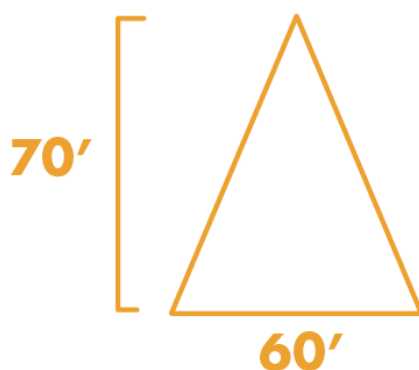
▲ **Triangle = (0.5 × base) × height**



$$\begin{aligned}\text{Area} &= 14' \times 14' \\ &= \mathbf{196 \text{ sq ft}}\end{aligned}$$



$$\begin{aligned}\text{Area} &= 17' \times 24' \\ &= \mathbf{408 \text{ sq ft}}\end{aligned}$$



$$\begin{aligned}\text{Area} &= (0.5 \times 60') \times 70' \\ &= (30') \times 70' \\ &= \mathbf{2,100 \text{ sq ft}}\end{aligned}$$

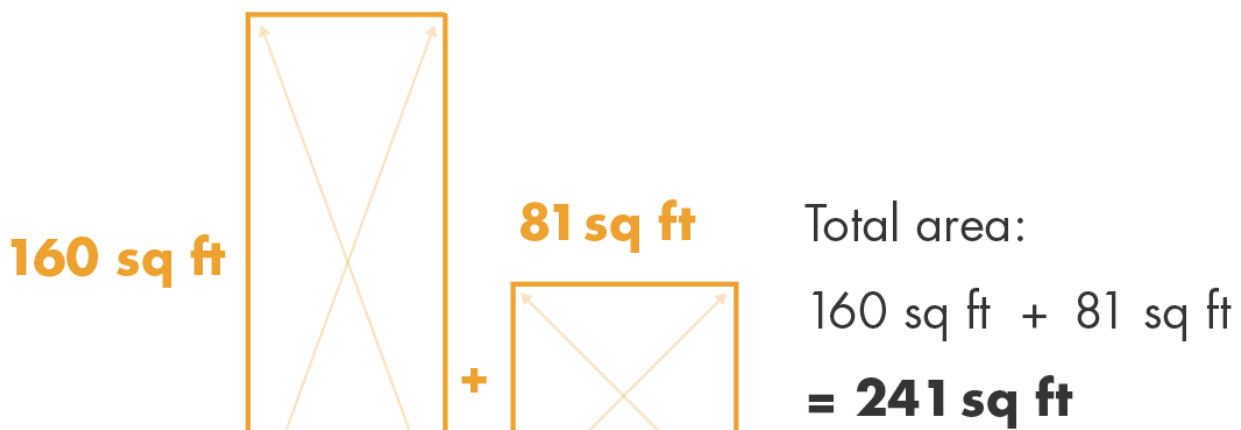
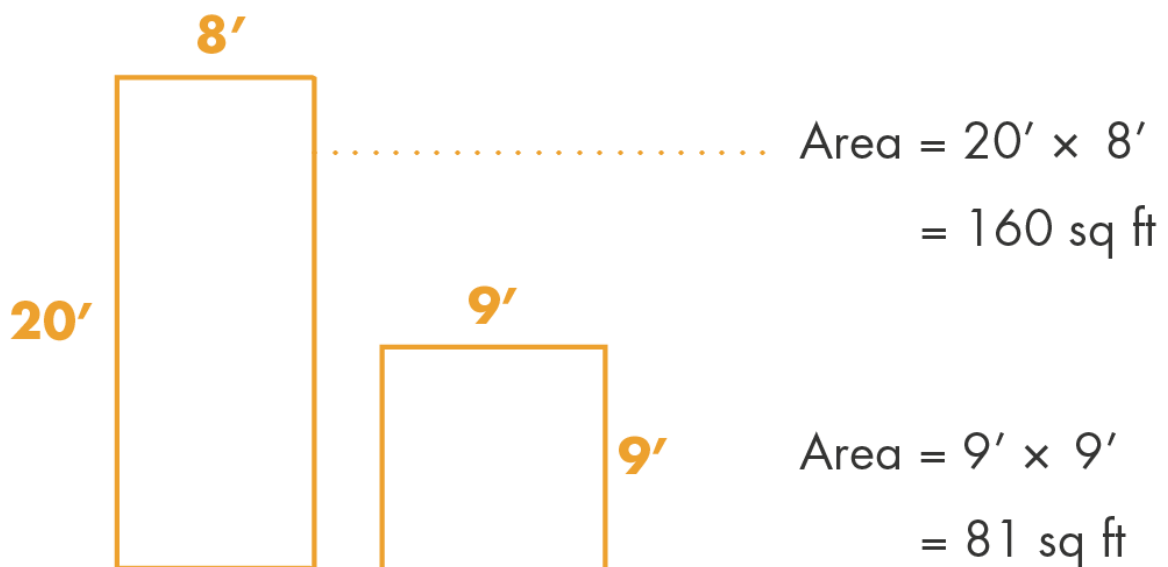
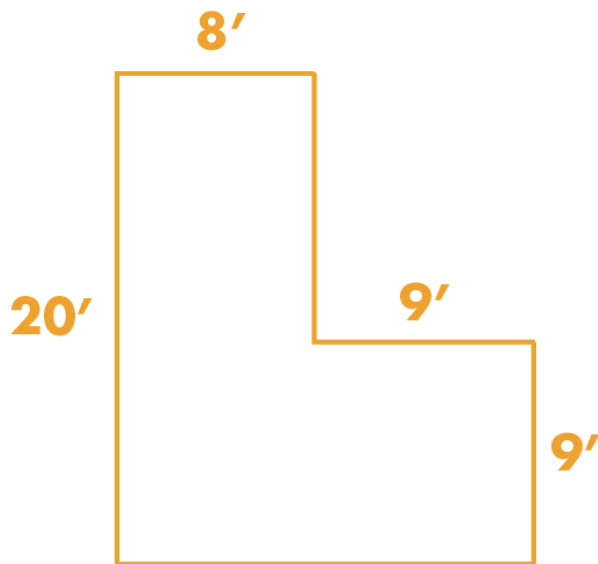
Find Area

■ **Irregular shapes:**

1. Divide the shape into squares, rectangles,

and triangles.

2. Find the area of each separate shape.
3. Add the area of the separate shapes together.





Acreage and Hectares

Acreage = property size (sq ft) ÷ 43,560 sq ft

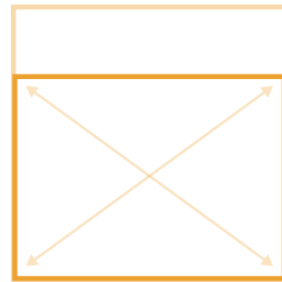
Hectares = property size (acres) ÷ 2.47 acres

A lot has an area of 30,000 square feet.

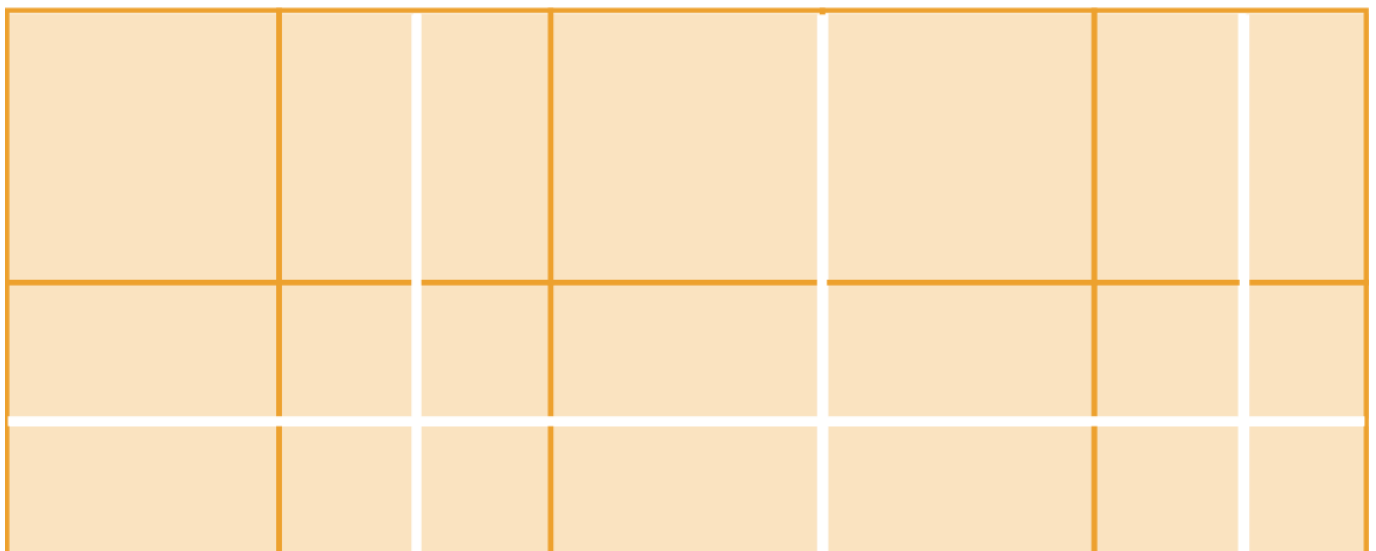
How many acres is this?

$$30,000 \text{ sq ft} \div 43,560 \text{ sq ft} = \mathbf{0.69 \text{ acres}}$$

1 acre



30,000 sq ft



1 hectare

1 acre

A lot has an area of 10 acres.

How many hectares is this?

$$10 \text{ acres} \div 2.47 \text{ acres} = \mathbf{4.05 \text{ hectares}}$$

Price per Square Foot

Price per square foot = Sales price $\div$ Area

A 2,100-square-foot home just sold for \$150,000.

What's the price per square foot?

$$\$150,000 \div 2,100 \text{ square feet} = \mathbf{\$71.43 \text{ per sq ft}}$$

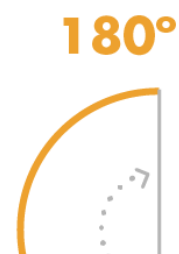
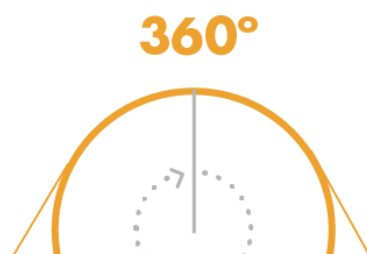
Circular Measures

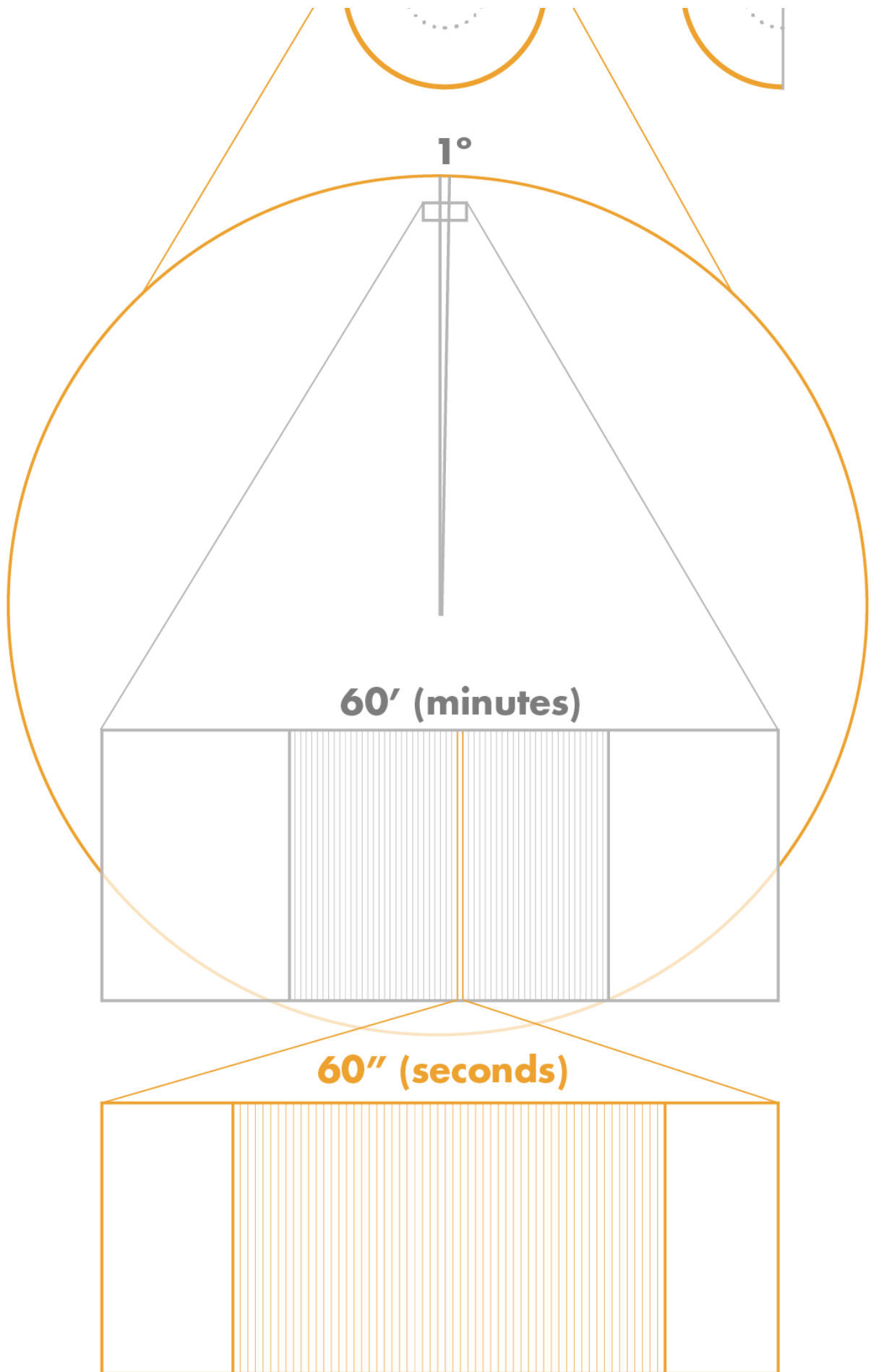
360 degrees = Circle

180 degrees = Semi-circle

1 degree = 60 minutes

1 minute = 60 seconds

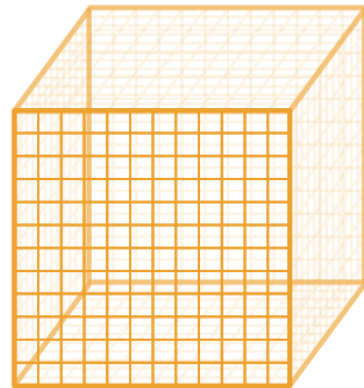




Cubic Measures

$$1,728 \text{ in}^3 = 1 \text{ ft}^3 (12 \times 12 \times 12 \text{ in})$$

$$27 \text{ ft}^3 = 1 \text{ yd}^3$$

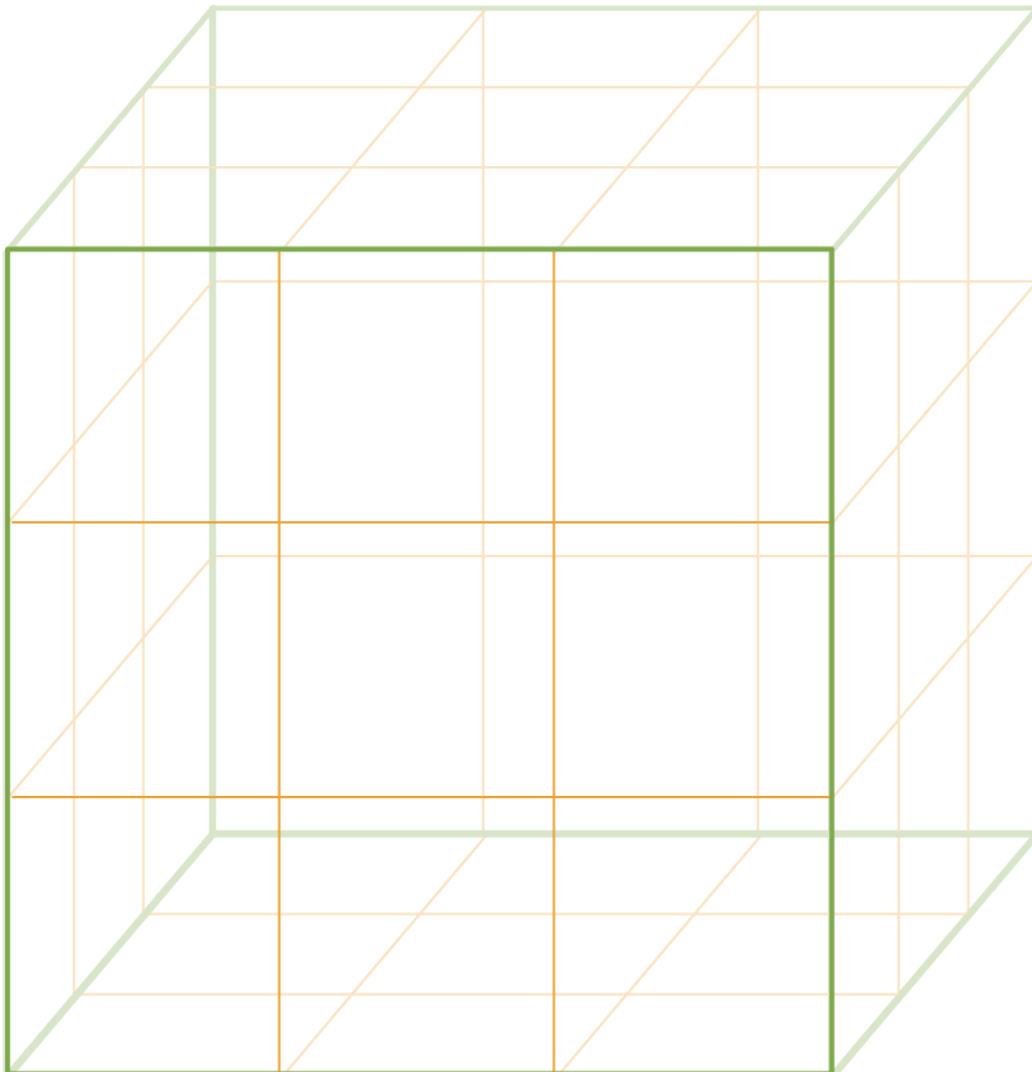


1 ft

← 12 in →



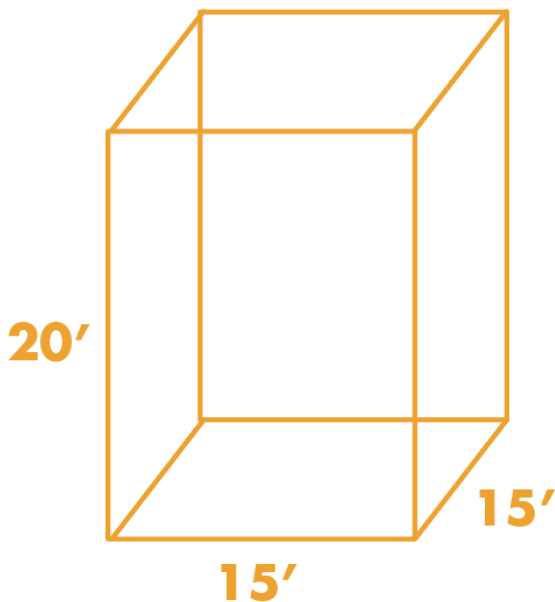
← **1 yd (3ft)** →



Volume

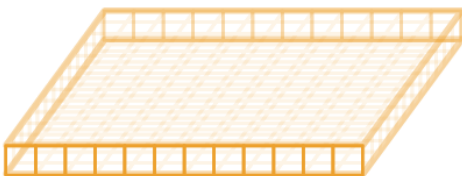
For most shapes, air space (or volume) is calculated by multiplying area times the height of the object.

**Cube or rectangular volume =
Length x Width x Height**



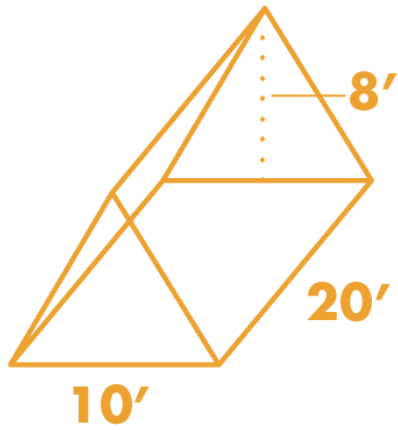
$$\begin{aligned}\text{Volume} &= 15' \times 15' \times 20' \\ &= \mathbf{4,500 \text{ cubic ft}}\end{aligned}$$

1 board foot = 144 cubic inches



$$\begin{aligned}\text{Volume} &= 12'' \times 12'' \times 1'' \\ &= \mathbf{144 \text{ cubic in}}\end{aligned}$$

**Triangular volume =
 $\frac{1}{2}$ (Length x Width x Height)**



$$\begin{aligned}\text{Volume} &= \frac{1}{2} (10' \times 20' \times 8') \\ &= \frac{1}{2} (1,600') \\ &= \mathbf{800 \text{ cubic ft}}\end{aligned}$$